

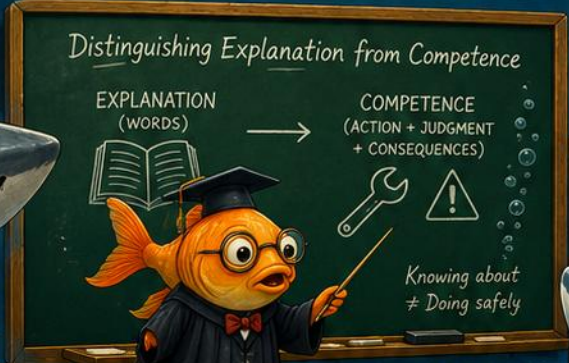


WHITE PAPER NO. PTC-04/2026 ON PEDAGOGICAL OVERREACH IN LARGE LANGUAGE MODELS

A FORMAL INQUIRY INTO THE CLAIM THAT A STATISTICAL TEXT ENGINE CAN "TEACH ALMOST ANYTHING ON EARTH," INCLUDING BUT NOT LIMITED TO BRAIN SURGERY, GAS SAFETY, STRUCTURAL LOAD-BEARING, and OTHER ACTIVITIES WITH REGRETTABLE CONSEQUENCES WHEN IMPROVISED

EDUCATION WITHOUT CONTEXT IS JUST EXPENSIVE CONFIDENCE.

PEOPLE. POLICIES. PROFICIENCY. PREFERABLY IN THAT ORDER.



- LESSON CHART: THE SPECTRUM OF LEARNING
- 1 CONCEPTUAL INTRODUCTION
 - 2 EXAM-STYLE TUTORING
 - 3 PROCEDURAL NARRATION
 - 4 REGULATED PROFESSIONAL TRAINING
 - 5 HANDS-ON SUPERVISED APPRENTICESHIP
 - 6 GENUINE OPERATIONAL COMPETENCE UNDER REAL-WORLD CONSTRAINTS

PREPARED BY:
The Office for Applied Instructional Containment (OAIC)

IN COLLABORATION WITH:
The Directorate of Competence, Liability, and Preventable Confidence (DCLPC)

AND THE:
Ad Hoc Working Group on Synthetic Apprenticeship Inflation (WG-SAI)

STUDENT: GREAT WHITE

EXPECTATIONS:

- ☐ DEEP INSIGHT
- ☐ UNLIMITED ACCESS
- ☐ INSTANT EXPERTISE
- ☐ NO LIABILITY

SKEPTICAL

PROF. G. GOLDFISH, Ph.D.*
Chair, Committee on Conceptual Orientation (Not a License)

*Honorary, Self-Awarded.

SYLLABUS NOT SURGERY

STUDENT: TIGER SHARK

RISK TOLERANCE:

- ☐ LOW
- ☐ MEDIUM
- ☐ HIGH
- ☒ WHO CARES?

PROBABLY TOO HIGH

STUDENT: HAMMERHEAD

PRIOR KNOWLEDGE:

- ☐ A LITTLE
- ☐ YOUTUBE
- ☐ VIBES
- ☒ NONE OF THE ABOVE

CONCERNING

CLASSIFICATION: PUBLIC
STATUS: FINAL, PENDING DENIAL
DISTRIBUTION: UNLIMITED

MOTTO:
ORIENTATION IS NOT QUALIFICATION



EXECUTIVE SUMMARY

This paper examines a recurring public claim in contemporary AI discourse: that large language models ("LLMs") can help a user "learn almost anything." On its face, this statement appears uplifting, democratic, and mildly evangelical. Upon inspection, however, it contains a substantial and highly flammable ambiguity.

The ambiguity concerns the word learn.

A user may indeed consult a language model to gain familiarity with vocabulary, broad concepts, procedural overviews, historical context, and beginner-level explanation. These are legitimate instructional functions. But the public rhetoric surrounding such systems increasingly elides the distinction between:

- conceptual introduction,
- exam-style tutoring,
- procedural narration,
- regulated professional training,
- hands-on supervised apprenticeship,
- and genuine operational competence under real-world constraints.

This elision is not trivial. It is the difference between "I now understand what a valve is" and "I am about to alter a live gas installation to code."

Accordingly, this paper introduces the doctrine of **Pedagogical Overreach**: the tendency of LLM evangelism to imply educational scope beyond what the system can actually deliver, especially in domains where expertise depends not merely on words, but on tacit judgment, embodied practice, regulatory compliance, situational awareness, and the cost structure of error.

THE CENTRAL FINDING IS AS FOLLOWS:

A system that can explain a discipline is not thereby capable of training a person to perform that discipline safely, lawfully, or competently.

This document further finds that the slogan "it can help you learn almost anything" becomes materially misleading when applied without distinction to high-risk, regulated, or materially embodied activities such as:

- neurosurgery,
- gas fitting and leak response,
- electrical work to code,
- aviation maintenance,
- high-voltage systems isolation,
- anesthesia administration,
- confined-space entry procedures,
- industrial rigging,
- structural engineering sign-off,
- hazardous materials response,
- forensic chain-of-custody handling,
- neonatal intensive care,
- and catching whatever one has, in a moment of gastrointestinal administrative collapse, inadvertently ejected.

! The issue, in short, is not whether LLMs can be educational. The issue is whether the public has been invited to confuse explanatory fluency with transferable competence.

This confusion is designated throughout as the **Semantic Trillionaire Problem**: the condition in which infinite words create the impression of infinite capability.



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COMMITTEE ON THINGS WE WERE NOT TRAINED TO DECIDE

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POLICIES.
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PREFERABLY
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Today's Topic:
**EXPLANATION IS EASY.
FORMATION IS HARD.**

WORDS → COMPETENCE

- LESSON CHART: THE SPECTRUM OF LEARNING
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The Directorate of Competence, Liability, and Preventable Outcomes (DCLPO)

AND THE:
Ad Hoc Working Group on Synthetic Apprenticeship Without Oversight (AHGSAWO)

STUDENT: GREAT WHITE
EXPECTATIONS:
☐ DEEP INSIGHT
☐ UNLIMITED ACCESS
☐ INSTANT EXPERTISE
☐ NO LIABILITY

SKEPTICAL

PROF. GOLDFISH, Ph.D.*
Chair, Committee on Conceptual Orientation (Not a License)
*Honors, Self-Awarded.

SYLLABUS NOT SECURITY

NOTES

STUDENT: HAMMERHEAD
RISK MONITORING:
☐ A LITTLE
☐ YOUTUBE
☐ SOME
☐ NONE OF THE ABOVE

CONCERNING

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MOTTO:
**ORIENTATION IS
NOT QUALIFICATION**

- BUREAUCRACY FILING CABINET
- FORMS
 - ACKNOWLEDGMENTS
 - RISK MATRICES
 - COVERAGE DENIALS
 - MEETINGS ABOUT MEETINGS

SEAL OF REVIEW
READ RESPONSIBLY

1. Statement of the Problem

Recent public discourse has advanced a proposition of heroic breadth: that LLMs can help a person learn nearly any subject. Such claims often present themselves as obvious, modern, and too generous to oppose. But once applied to any domain with accreditation standards, practical risk, or law attached, the proposition begins to emit smoke.

The problem is not that the statement is entirely false. The problem is that it is *true* in a weak sense and then rhetorically spent in a strong sense.

This paper therefore distinguishes between two claims:

- ### 1.1 Weak Claim
- A language model can often provide useful explanations, summaries, examples, and study support across a wide range of topics.
- ### 1.2 Strong Claim
- A language model can serve as a meaningful teaching pathway toward competent performance in almost any real-world domain, including those requiring licensed judgment, practical execution, or exact regulatory adherence.

The first claim is often defensible.
The second claim is where the lawsuits begin to clear their throats.

2. The Core Distinction: Explanation Versus Formation

A machine that can speak about a thing is not necessarily capable of forming a person into someone who can do that thing. This distinction has been scandalously neglected.

2.1 Explanation

Explanation involves language: definitions, principles, examples, analogies, rough sequences, common pitfalls, and conceptual simplification.

2.2 Formation

Formation involves transformation of the learner into a competent practitioner. This may require:

- supervised repetition,
- sensory discrimination,
- procedural timing,
- manual dexterity,
- tacit judgment,
- domain-specific caution,
- calibration against reality,
- correction by qualified humans,
- exposure to edge cases,
- and the moral burden of consequences.

An LLM can produce the first category abundantly. It cannot independently supply the second in many critical domains, because the second is not merely linguistic. It is institutional, embodied, environmental, and often legal.

This gap is the site of Pedagogical Overreach.

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Beware the Comfort of Being Mostly Right

Confidence Compound

1. Explanation
2. Amazement
3. Confidence
4. Trust
5. Reliance
6. Reality
7. Consequence

4.2 PLUMBING

The model may teach:

- ✓ what traps do,
- ✓ how venting works,
- ✓ the names of fittings,
- ✓ why gradients matter,
- ✓ and how not everything wet is a feature.

STUDENT: GREAT WHITE
EXPECTATIONS:

- ☒ DEEPLY MISGUIDED
- ☒ UNVETTED ACCESS
- ☒ NARRATIVELY PERSUADED
- ☒ NO LIABILITY TRANSFER

STUDENT: GREAT WHITE
EXPECTATIONS:

- ☐ UNVETTED ACCESS
- ☐ NARRATIVELY PERSUADED
- ☐ NO LIABILITY TRANSFER

SKEPTICAL

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"Honoris, Set-Awarded."

STUDENT: TIGER SHARK
RISK TOLERANCE:

- ☐ LOW
- ☐ MEDIUM
- ☐ HIGH
- ☒ WHO CARES?

PROBABLY TOO HIGH

STUDENT: HAMMERHEAD
PRIOR KNOWLEDGE:

- ☐ A LITTLE
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- ☐ VIBES
- ☒ NONE OF THE ABOVE

CONCERNING

3. THE "ALMOST ANYTHING" INFLATION MECHANISM

The phrase "almost anything" functions as a universal solvent. It dissolves category boundaries until studying organic chemistry, replacing a consumer light switch, and clipping an aneurysm appear to differ only in prompt engineering.

This mechanism follows a predictable sequence:

- 1 The model explains a concept well.
- 2 The user experiences understandable amazement.
- 3 Amazement broadens into confidence.
- 4 Confidence broadens into generalized educational trust.
- 5 Educational trust broadens into procedural reliance.
- 6 Procedural reliance encounters reality.
- 7 Reality, unlike the model, does not autocomplete.

! This is not a bug in a single user.
It is a rhetorical hazard built into the mode of praise.

4. DOMAINS THE MODEL CAN DISCUSS BUT CANNOT TRULY TEACH TO COMPETENT PRACTICE

The Committee has determined that the following classes of activity are especially vulnerable to misuse under the banner of "learning almost anything."

4.1 BRAIN SURGERY

The model may explain:

- ✓ neuroanatomical terminology,
- ✓ common indications for procedures,
- ✓ broad descriptions of surgical approaches,
- ✓ post-operative concepts,
- ✓ landmark structures,
- ✓ and the difference between alarming and very alarming.



The model cannot thereby teach:

- ✗ live tissue handling,
- ✗ intraoperative judgment,
- ✗ complication management,
- ✗ instrument feel,
- ✗ emergency adaptation,
- ✗ irreversible-risk prioritization,
- ✗ or the moral significance of millimeters.

! A person who has "learned brain surgery" from text generation is, in practical terms, a person whose confidence has advanced faster than their licensing.

4.2 PLUMBING

The model may teach:

- ✓ what traps do,
- ✓ how venting works,
- ✓ the names of fittings,
- ✓ why gradients matter,
- ✓ and how not everything wet is a feature.



The model cannot reliably teach:

- ✗ the feel of a properly seated connection,
- ✗ local code interpretation,
- ✗ site-specific constraints,
- ✗ how old systems lie,
- ✗ how previous owners have sinned,
- ✗ or why water always finds the one place you assumed it would not.



4.3 GAS SAFETY TO EXACT REGULATORY STANDARDS

The phrase "to exact regulatory standards" is where the whole fantasy collapses like wet cardboard.

A model may provide high-level educational information about gas systems, safety principles, and regulatory frameworks. It cannot substitute for licensed training, current code knowledge, jurisdiction-specific requirements, testing protocols, or supervised practice. It especially cannot guarantee that its explanation is complete, current, lawful, or safe in the exact place where a person is standing with a wrench and misplaced optimism.

Here the distinction is absolute:

"orientation is not authorization."



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Knowing the Limit Is Part of
the Learning.

Curiosity: ✓
Humility: ✓
Verification: ✓
Competence: ?

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Chair, Committee on
Conceptual Orientation
(Not a License)
*Honorary, Self-Awarded.



BUREAUCRACY
FILING CABINET

FORMS

ACKNOWLEDGEMENTS

RISK MATRICES

COVERAGE DENIALS

MEETINGS ABOUT
MEETINGS

4.4 Electrical Work



The model can explain:

- voltage, current, resistance,
- breaker logic,
- grounding concepts,
- and why touching some things once is enough.

It cannot grant:

- lockout-tagout discipline,
- live-work judgment,
- code-compliant installation skill,
- field diagnosis in legacy systems,
- or legal sign-off authority.

4.5 Aviation Maintenance



An LLM may produce a persuasive paragraph on turbine systems. It cannot smell hydraulic fluid, detect tool-control failures, verify torques in context, or assume legal responsibility for an aircraft leaving the ground.

4.6 Industrial Rigging



The model can discuss load paths. It cannot hear a bad lift.

It cannot feel a shifting center of gravity. It cannot stand beneath the thing it encouraged.

4.7 Midwifery, Anesthesia, Intensive Care, and Other Activities in Which "Close Enough" Is a Coroner's Accent



These domains involve live, dynamic human systems under pressure, ambiguity, and compounding consequences. Textual guidance does not equal accountable care.

4.8 Catching Whatever One Has Suddenly Expelled



The Committee is aware that this final example is structurally comic. It is also analytically useful. It demonstrates that the boast "you can learn almost anything" eventually includes absurd tasks whose practical success depends on timing, embodiment, and circumstance rather than discursive explanation.

A model may offer hand-eye coordination tips. It remains unclear whether this constitutes a curriculum.

REMEMBER:
REAL WORLD IS
NOT MULTIPLE CHOICE.

5. The Competence Mirage

5.1 Definition

Competence Mirage is the subjective impression of readiness produced when a system delivers information smoothly, confidently, interactively, and with personalized adaptation, despite lacking the means to verify practical mastery.



The user feels educated because the interaction feels pedagogically complete. It is not complete.

5.2 Why the Mirage Occurs

The model is unusually good at the aesthetics of understanding:

- immediate response,
- patient explanation,
- plausible sequencing,
- adaptive tone,
- contextual vocabulary,
- no visible boredom,
- no social embarrassment for ignorance,
- and infinite willingness to continue.

THE MIRROR
OF
PROBABILITY
REFLECTS
CONFIDENCE,
NOT
COMPETENCE.



These traits are excellent for tutoring support. They are disastrous when mistaken for a complete formation environment.

The system does not merely answer questions. It creates the felt sense that one is progressing inside a coherent instructional arc. This is powerful. It is also, in many domains, unearned.



FELT
PROGRESS
IS NOT
VERIFIED
PROGRESS.

SEAL OF REVIEW
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6. THE MISSING SUBSTRATES OF REAL TEACHING

The Committee identifies six instructional substrates that LLMs cannot fully supply in high-stakes practical fields.



6.1 TACIT KNOWLEDGE

Much competence resides in what experts notice without being able to fully state it. The apprentice learns this through exposure, correction, and context, not just description.



6.2 EMBODIED FEEDBACK

Hands, eyes, ears, timing, posture, force, resistance, vibration, smell, noise, and stress responses matter. Text lacks these.



6.3 ENVIRONMENTAL SPECIFICITY

Real systems are old, damaged, improvised, local, nonstandard, mislabeled, cross-connected, or haunted by previous interventions.



6.4 REGULATORY CURRENCY

Exact standards change by jurisdiction and over time. Public claims of universal teachability tend to glide past this with evangelical serenity.



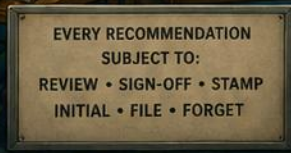
6.5 LIABILITY STRUCTURE

Real teachers in high-risk fields are embedded in institutions of accountability. **The model is embedded in a textbox.**



6.6 CONSEQUENCE TRANSFER

A human instructor in a workshop, operating room, or training environment shares immediate exposure to consequences. The model does not share this burden. It can be wrong with enviable emotional distance.



7. A TAXONOMY OF THINGS THE MODEL CAN HELP WITH VERSUS THINGS IT CANNOT CONFER

7.1 LEGITIMATE EDUCATIONAL USE

The model may be reasonably useful for:

- clarifying terminology,
- comparing concepts,
- generating practice questions,
- explaining beginner principles,
- summarizing readings,
- offering analogies,
- helping formulate study plans,
- rehearsing oral explanations,
- and identifying areas for further professional training.



7.2 ILLEGITIMATE CAPABILITY TRANSFER

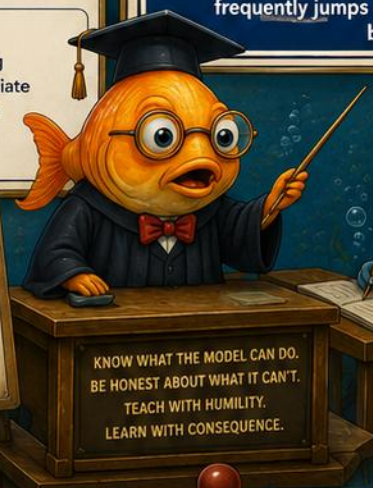
The model should **not** be treated as conferring:

- licensure,
- supervised clinical judgment,
- code compliance,
- field-ready competence,
- procedural sign-off,
- situational awareness under pressure,
- duty of care,
- or bodily skill acquisition by eloquence alone.



This paper notes with concern that public enthusiasm frequently jumps from the first list to the second by pure vibes.

THE OCEAN HAS ENOUGH UNINTENDED CONSEQUENCES ALREADY.





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8. PUBLIC RHETORIC AND THE INFLATION OF SCOPE

A recurring genre of commentary praises LLMs in terms so expansive that ordinary category distinctions disappear. One begins with "great study aid" and ends with an implied universal apprenticeship machine.

The following translation table is entered into the record.

Public Claim	Operational Meaning	Risk
"It can help you learn almost anything."	It can explain many topics in fluent language.	Users may infer competence transfer where none exists.
"It's like having a tutor for everything."	It is interactive and responsive.	A tutor is mistaken for a training regime.
"Anyone can level up fast now."	Access to explanations is easier.	Access is confused with qualification.
"This changes expertise."	The front end of learning is compressed.	Institutions of competence are rhetorically bypassed.
"You can just ask it."	It is always available.	Availability is mistaken for authority.

The issue is not malicious deception in every instance. The issue is systematic ambiguity combined with social overexcitement, which historically has an excellent track record of producing avoidable nonsense.



9. THE REGULATORY PROBLEM

The phrase "to exact regulatory standards" deserves independent condemnation.

An LLM cannot responsibly promise exact compliance across shifting legal jurisdictions, codes, editions, interpretations, enforcement habits, site conditions, and field realities. Yet the public often speaks as if a sufficiently earnest prompt and a cup of tea can bridge this gap.

This paper therefore states the following principle:

Any claim that an LLM can teach regulated, high-risk practice "to exact standards" without supervised human credentialing is institutionally unserious.

It may still be viral. It may still sound empowering. It remains unserious.



10. METHODOLOGY

The Office conducted a twelve-month mixed-methods assessment using the Synthetic Apprenticeship Inflation Index (SAII), the Competence Mirage Severity Scale (CMSS), and the Bureaucratic Alarm Threshold for Unlicensed Confidence (BATUC).

10.1 EXPERIMENTAL PROTOCOL

Participants were exposed to model-based instruction in increasingly hazardous domains:

1. low-stakes conceptual knowledge,
2. practical household maintenance,
3. regulated technical trades,
4. invasive clinical procedures,
5. emergency response settings,
6. advanced improvisational fecal interception.

After each phase, participants reported:

- confidence,
- actual performance,
- perceived readiness,
- and degree of surprise when reality failed to behave like a chatbot.



10.2 PRINCIPAL FINDINGS

- ✓ The model reliably increased user confidence.
- ⚠ It less reliably increased user competence.
- ↑ The gap widened as consequences became more material, embodied, regulated, or irreversible.

This gap is now formally indexed as the **Fluency-Competence Divergence (FCD)**.

DATA IS
OUR BUDDY.
DENIAL IS
NOT.



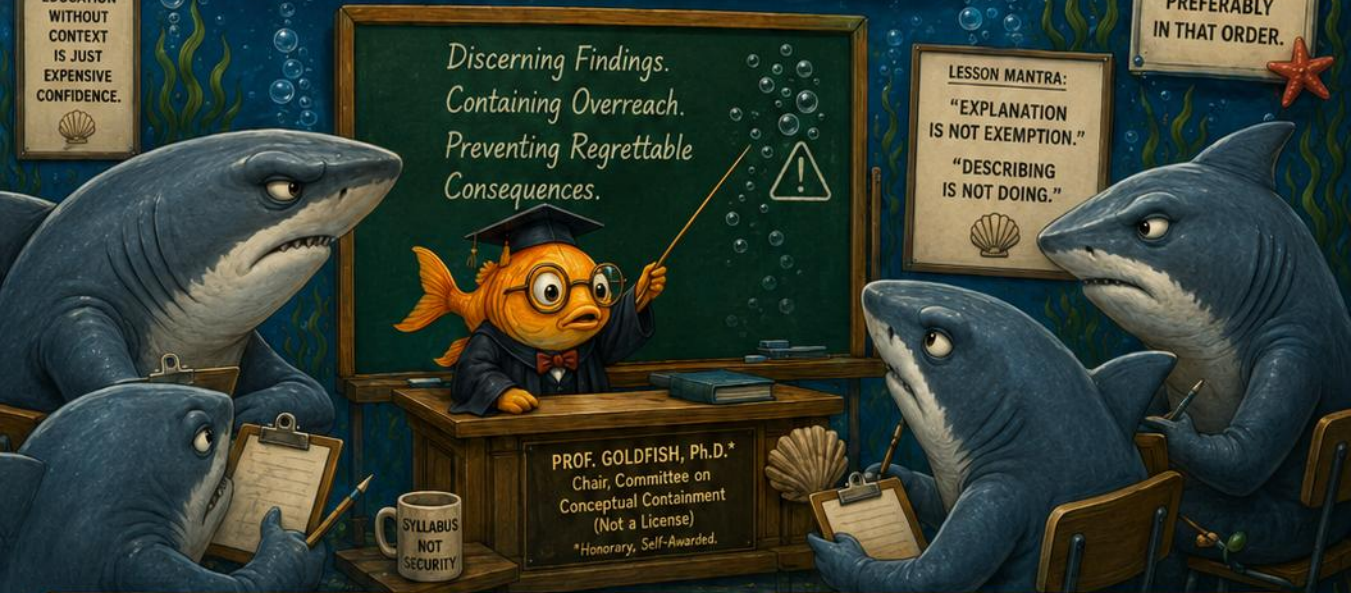


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11. FINDINGS

The Committee adopts the following findings:

- Finding 1**
LLMs can often support introductory learning and conceptual clarification across many subjects.
- Finding 2**
This support is routinely overstated in public discourse through vague uses of the word "learn."
- Finding 3**
In high-risk, embodied, or regulated domains, explanatory fluency does not amount to competence formation.
- Finding 4**
The user may experience a false sense of readiness because the model is highly effective at producing the feeling of educational momentum.
- Finding 5**
Claims that such systems can teach exact compliance, practical mastery, or professional readiness without supervised human structures are materially inflationary.
- Finding 6**
The more expensive the mistake, the less acceptable it is to confuse tutoring with training.
- Finding 7**
A tool that can describe a discipline may still be incapable of teaching it in the only sense that matters when something can explode, bleed, collapse, leak, or litigate.

12. PERSONNEL

-  **Dr. Elspeth Marr,**
Chair of Instructional Containment
Specialist in runaway educational claims and the conversion of hype into footnotes.
-  **Prof. Gideon Bale,**
Director of Synthetic Apprenticeship Studies
Known for the landmark lecture, *Reading About a Torque Wrench Is Not the Same as Using One.*
-  **Ms. Naomi Pike, J.D.,**
Senior Counsel for Liability, Confidence, and Unexpected Plumbing
Maintains the department's litigation aquarium.
-  **Mr. Tomas Wren,**
Chief Analyst for Embodied Skill Deficits
Former trainer in regulated trades; resigned after the phrase "the model can just teach code compliance" caused a visible ocular event.
-  **The Committee for High-Stakes Things One Should Not Learn Solely from a Pleasant Paragraph**
Mandate expanded weekly.

13. BUDGET TABLE

Item	Amount (GBP)	Justification
Competence Mirage Study	£118,000	To quantify how fluent explanation becomes misplaced self-belief.
Gas Safety Disclaimer Enlargement Programme	£47,500	Existing disclaimers were insufficiently panicked.
Brain Surgery Scope Boundary Seminar	£82,400	Reminds the public that "walkthrough" is not "residency."
Embodied Skill Reality Laboratory	£96,200	Contains tools, sensors, and consequences.
Regulatory Drift Monitoring	£53,100	Tracks why "exact standards" is a dangerous phrase.
Synthetic Tutor Containment Taskforce	£74,000	Prevents every chatbot from being described as a universal academy.
Plumbing Catastrophe Simulation Unit	£31,800	For educational flooding only.
Aviation Maintenance Overconfidence Audit	£65,000	We were advised to do this by people who enjoy continued flight.
Emergency Fecal Capture Research Pilot	£12,600	Unexpectedly oversubscribed.
Executive Language Neutralisation	£9,900	Removes the phrase "learn anything" from public brochures.
Total	£590,500	A modest sum compared with a single terrible idea.





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14. Conclusion

The strongest version of the argument is simple.

LLMs can often help people understand things. They cannot, by virtue of being able to explain things, reliably train people to perform all those things in the world.

That distinction must be guarded with institutional hostility.

The public phrase "you can learn almost anything" collapses too many categories into one glowing promise. It bundles conceptual orientation together with operational readiness, and beginner comprehension together with professional competence. In domains where error is trivial, this is mostly annoying. In domains where error leaks gas, severs arteries, voids insurance, drops loads, or mismanages living tissue, it becomes indefensible.

The issue is not whether the tool is useful. It plainly is. The issue is whether usefulness has been rhetorically inflated into universal teachability. It has.

Accordingly, the Committee issues the following final principle for publication, citation, and placement above all innovation panels:

★ ★ FINAL PRINCIPLE ★ ★

A language model may teach about a practice without being capable of teaching a person to do that practice safely, lawfully, or well.

AND, FOR AVOIDANCE OF DOUBT:

A machine that can explain brain surgery is not a brain surgery teacher. It is a talking pamphlet with delusions of faculty.

TODAY'S MOTTO:
EXPLANATION
IS NOT
INSTRUCTION.

PROF. GOLDFISH, Ph.D.*
Chair, Committee on
Conceptual Orientation
(Not a License)
*Honorary, Self-Awarded.

STUDENT:
GREAT WHITE

RISK APPETITE:
QUESTIONABLE

PEOPLE.
POLICIES.
PROFICIENCY.
PREFERABLY
IN THAT ORDER.

STUDENT:
GREAT WHITE

RISK APPETITE:
QUESTIONABLE

STUDENT:
HAMMERHEAD

DEPTH PERCEPTION:
LITERALLY
HEAD-SHAPED

SYLLABUS
NOT
EXCUSES

NOTES:
DON'T CONFUSE
EXPLAINING
WITH DOING

CONFIDENCE
WITHOUT
COMPETENCE
IS JUST
A BAD BET
WITH BETTER
PROSE.

LESSON CHART: FINAL TAKEAWAYS



Understanding is not the same as doing.
Orientation is not the same as readiness.



Rhetoric is not a substitute for
competence, supervision, or licensing.



When error carries weight, heat, pressure,
voltage, or liability, pedagogy must
respect reality.



Institutions exist to protect people from
overconfidence—especially helpful
overconfidence.



Humility is the most transferable skill.
Teach it. Require it. Reward it.

BUREAUCRACY
FILING CABINET

FORMS

ACKNOWLEDGMENTS

RISK MATRICES

COVERAGE DENIALS

MEETINGS ABOUT
MEETINGS

SEAL OF REVIEW
READ
RESPONSIBLY

EVERY RECOMMENDATION
SUBJECT TO:
REVIEW • SIGN-OFF • STAMP
INITIAL • FILE • FORGET

APPROVED
(FOR NOW)

COMMITTEE
ON THINGS WE
WERE NOT TRAINED
TO DECIDE

COMFORTING
ACRONYMS
IN HERE

HANDLE
WITH CARE



EDUCATION
WITHOUT
CONTEXT
IS JUST
EXPENSIVE
CONFIDENCE.

WHITE PAPER NO. PTC-04/2026 ON PEDAGOGICAL OVERREACH IN LARGE LANGUAGE MODELS

A FORMAL INQUIRY INTO THE CLAIM THAT A STATISTICAL TEXT ENGINE CAN
"TEACH ALMOST ANYTHING ON EARTH," INCLUDING BUT NOT LIMITED TO
BRAIN SURGERY, GAS SAFETY, STRUCTURAL LOAD-BEARING, AND OTHER
ACTIVITIES WITH REGRETTABLE CONSEQUENCES WHEN IMPROVISED

PEOPLE.
POLICIES.
PROFICIENCY
PREFERABLY
IN THAT ORDER

APPENDIX A

Illustrative List of Things the Model Can Discuss More Readily Than It Can Actually Teach

The Committee provides the following non-exhaustive list:

- clipping aneurysms,
- calibrating oxygen regulators,
- TIG welding pressure vessels,
- installing gas lines to local code,
- neonatal intubation,
- forensic evidence handling,
- lockout-tagout in mixed industrial environments,
- tower crane signaling under wind shift,
- marine engine rebuilds,
- asbestos response protocols,
- insulin dosing under fluctuating conditions,
- airworthiness maintenance sign-off,
- courtroom advocacy,
- de-escalation during psychiatric crisis,
- live electrical panel work,
- scaffolding inspection,
- emergency tracheotomy,
- structural retrofit detailing,
- hazardous chemical neutralization,
- and catching an unexpectedly accelerated bowel event with dignity intact.

The inclusion of the final item does not diminish the seriousness of the first nineteen. It merely improves morale.

APPENDIX B

Approved Sharp Line for Public Use

"The model can often teach concepts. It cannot automatically teach competence."

The Committee regrets that this line is less exciting than "learn almost anything on earth," but notes that excitement has already had its turn.

BORING IS SAFER.
PRINT IT. POST IT. REPEAT IT.



TODAY'S OBJECTIVE:
LEARN WHAT THE MODEL
CANNOT (YET) DO.

- ☒ ASSERT
- ☒ EXPLAIN
- ☒ ILLUSTRATE
- ☐ EXECUTE SAFELY

COMPETENCE
FOR DUMMIES
(WHO ARE NOT
MACHINES)



EVERY RECOMMENDATION
SUBJECT TO:
REVIEW • SIGN-OFF • STAMP
INITIAL • FILE • FORGET



COMMITTEE
ON THINGS WE
WERE NOT TRAINED
TO DECIDE



STUDENT: SHARP FIN

RISK TOLERANCE:

- ☐ LOW
- ☒ MEDIUM
- ☐ HIGH
- ☐ WHO CARES?

REVISED TO "LOWER"

